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Effects of Cannabidiol Ingestion on Thermoregulatory and Inflammatory Responses to Treadmill Exercise in the Heat in Recreationally Active Males

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ABSTRACT

Purpose: Exertional heat stress can induce systemic endotoxin exposure and a pro-inflammatory cascade, likely impairing thermoregulation. Cannabidiol (CBD) is protective in pre-clinical models of tissue ischaemia and inflammation. Therefore, this study examined the effects of CBD ingestion on exercise-induced thermoregulatory and inflammatory responses. **Methods:** In a randomised, double-blinded study, thirteen active males (age 25 ± 5 y; peak oxygen uptake [$\dot{V}O_2$peak] 50.4 ± 3.2 mL/kg/min) ingested 298 mg CBD or placebo 105 minutes before 1 h treadmill exercise (60-65% $\dot{V}O_2$peak) in 32°C and 50% relative humidity. Core temperature, skin temperature, heart rate, subjective outcomes and sweat loss were assessed during/after exercise. Plasma osmolality, plasma volume changes and plasma markers of intestinal damage (I-FABP), monocyte activation (CD14) and inflammatory cytokine responses (IL-6, IL-8 and TNF-α) were assessed at baseline, pre-exercise, 20- and 90-min post-exercise. **Results:** Core temperature (Δ 1.69 ± 0.48°C [CBD] and 1.79 ± 0.53°C [Placebo]) and I-FABP increased during exercise, with no differences between conditions ($p > 0.050$). Mean (95% CI) CD14 was 1776 (463 to 3090) pg/mL greater 90 min post-exercise in placebo ($p = 0.049$). Median (interquartile range) peak IL-6 concentration was -0.8 (-1.1, -0.3) pg/mL less in CBD ($p = 0.050$), whilst the between-conditions difference in IL-6 area under curve was -113 (-172, 27) pg/mL·270min ($p = 0.054$). **Conclusions:** CBD did not affect thermoregulation during exertional heat stress but appeared to elicit minor immunosuppressive effects, reducing CD14 and IL-6 responses, warranting investigation in humans under more severe heat strain and other pro-inflammatory scenarios.

Key Words: ACTIVITY, CANNABINOID, CANNABIS, HYDRATION, IMMUNE, RUNNING

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INTRODUCTION

Exertion in a hot environment, often accompanied by hypohydration, presents a thermoregulatory challenge (1,2). If physiological mechanisms cannot compensate for this additional heat stress, body temperature will rise, leading to impaired physical function, followed by exertional heat-illness/stroke, and in the most extreme cases, death (1). An endotoxin-induced inflammatory cascade has emerged as a central factor in this sequence of events, likely impairing thermoregulation and contributing to the adverse effects of heat illness (1,3,4). Modification of the external environment (e.g., weather or clothing) or reduction in exercise intensity is not always possible (e.g., in occupational settings). Therefore, there is growing interest in the potential for nutritional interventions to improve thermoregulation or mitigate the inflammatory response to exertional heat stress (5–7).

A nutritional intervention could target the prevention of intestinal damage and subsequent endotoxin release, or the inflammatory response resulting from endotoxemia. Damage to the intestinal epithelium often occurs during sustained physical exertion, likely via intestinal ischaemia, and is exacerbated by hypohydration and environmental heat stress (7,8). Intestinal damage is commonly assessed via serum/plasma intestinal-fatty acid binding protein (I-FABP), a protein involved in lipid metabolism and specific to the upper intestine that appears in circulation when the integrity of enterocytes is compromised (7,9). Following intestinal barrier disruption, bacteria move from the intestinal lumen into circulation, increasing blood lipopolysaccharide (LPS) concentrations (an endotoxin, and major component of the outer membrane of gram-negative bacteria) (10,11). LPS activates monocytes and neutrophils via toll-like receptor-4 (TLR4), initiating the secretion of inflammatory cytokines (e.g., interleukin-6 [IL-6], interleukin-

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8 [IL-8] and tumour necrosis factor-α [TNF-α]), which, when excessive, are accompanied by a fever and compromised thermoregulation (12–15). Cluster of differentiation-14 (CD14) acts as a co-receptor for LPS on monocyte TLR4 and is released by activated monocytes, meaning that elevated systemic soluble CD14 is indicative of LPS exposure in this context (7,16).

One potential preventative supplement that warrants investigation is cannabidiol (CBD), a non-intoxicating phytocannabinoid used for the treatment of epileptic seizures (Epidyolex®). CBD has gained substantial research interest for numerous potential therapeutic benefits, including anxiolytic, and antioxidant effects (17), and is used by recreational exercisers (18) and athletes (19). CBD has shown protective effects against damage to various tissues (e.g., neural, hepatic, renal and myocardial) during animal models of ischaemia (20–23). Acute ingestion of 600 mg CBD also attenuated aspirin-induced intestinal hyperpermeability in healthy adults (24). Furthermore, CBD reduced TLR4 signalling, subsequent activation of inflammatory signalling pathways, and secretion of IL-6, IL-8 and TNF-α across numerous *in vitro* models involving LPS stimulation (25–29). Several *in vivo* animal models have also shown that CBD doses of 1-80 mg/kg injected intraperitoneally, intravenously, or subcutaneously, reduced blood TNF-α and IL-6 responses after injection or intra-nasal instillation of LPS (30,31). Therefore, CBD administration could mitigate either the intestinal damage or subsequent inflammatory response induced by exertional heat stress (40).

There is little research examining the effect of CBD ingestion on body temperature in humans, although rodent studies have generally shown null effects (reviewed by McCartney et

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al.) (32). Some have found acute hypothermic effects in rats (33,34), whereas others have found hyperthermic effects in mice (35), and reasons for these discrepancies are unclear but could relate to doses used. However, no studies have examined thermoregulatory effects of CBD under conditions of exertional heat stress, during which CBD could interact with gastrointestinal disruption, endotoxaemia and inflammatory responses. Systemic elevations in LPS and inflammatory cytokines can increase core temperature (12,13), which is a critical determinant of exercise performance and heat illness risk (1). Therefore, the primary aim of this study was to examine the acute effects of cannabidiol ingestion on thermoregulation during exercise in a hot environment, with the secondary aims of examining the intestinal damage, CD14 and inflammatory cytokine responses to this exercise bout. It was hypothesised that CBD may attenuate the rise in core temperature during exercise, and attenuate increases in plasma I-FABP, CD14 and inflammatory cytokines.

METHODS

Participants and Study Design

Thirteen healthy, recreationally active, non-heat acclimated males (age 25 ± 5 y; body mass 79.5 ± 8.9 kg; body mass index 25.0 ± 2.6 kg/m^2; $\dot{V}O_2$peak 50.4 ± 3.2 mL/kg/min) completed two conditions (CBD vs placebo) in a double-blind, randomised, counterbalanced, crossover study design. Randomisation was performed via coin toss by a researcher external to data collection and analysis. At least one week separated conditions. In each condition, participants completed an exercise protocol in a hot environment, to examine thermoregulatory responses and effects on markers of intestinal damage and inflammation.

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A sample size estimation was performed using GPower 3.1 and data from Funnell et al. (36), which indicated that ten participants were required to detect a 0.4°C difference in mean core body temperature between conditions using a paired *t*-test (d_z = 1.0, repeated measures correlation = 0.51). Exclusion criteria were: aged <18 or >45 years, reported suffering from any blood-borne illness or morbidity that might influence cardiovascular or thermo-regulatory function (e.g., liver/renal/cardiovascular disease) or running ability (e.g., musculoskeletal injury), allergies to any products used in the study, use of cannabis or any CBD-containing product within two months of screening, or currently taking any prescription or over-the-counter medication. Participants were asked to report any supplement use at screening; whey protein, creatine and multivitamin use was reported. Participants were instructed to continue habitual supplement use and to not take any new supplements for the duration of their participation. All participants were recreationally active (completing at least 90 minutes of moderate or vigorous physical activity per week for at least three months before screening). Females were eligible to participate, but the full sample was recruited before any volunteered. Ethical approval was granted by the Loughborough University Ethics Approvals (Human Participants) Sub-Committee (LEON 8500).

Preliminary Visit

Participants attended the laboratory at least three days before main study visits, for assessment of peak oxygen uptake ($\dot{V}O_2$peak) and familiarisation with treadmill exercise (HP Cosmos Quasar, Nußdorf a. Inn, Germany) in an environmental chamber (Weiss Technik UK ltd, Loughborough, UK) at 32°C and 50% relative humidity (rh). Height and body mass were first recorded using a stadiometer (Seca, Hamburg, Germany) and scales (CFW-150, Adam

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Equipment, Milton Keynes, UK), respectively. $\dot{V}O_2$peak was then assessed using a modified step protocol in ambient conditions. Four four-minute stages were performed at increasing treadmill speeds (0% gradient), with $\dot{V}O_2$ assessed in the final minute of each stage using indirect calorimetry via the Douglas bag method. After this, treadmill speed was clamped and the gradient increased by 1% every minute until volitional fatigue, with $\dot{V}O_2$peak assessed during the final minute. Linear regression was used to determine treadmill speed at 60-65% $\dot{V}O_2$peak (8.4 ± 0.8 km/hour), at which participants then completed 20 minutes exercise in 32°C, 50% rh, with a front facing fan (wind speed ~4.7 km/h).

Expired gas samples were collected into Douglas bags via a mouthpiece and tubing, then analysed for O_2 and CO_2 concentrations via paramagnetic and infrared transducers, respectively (Mini HF 5200, Servomex Group Ltd., Surrey, UK). Ambient air was sampled proximally to participants during each expired gas sample to adjust for inspired O_2 and CO_2 concentrations (37). Volumes and temperatures of expired gas samples were measured using a dry gas meter (Harvard Apparatus, Kent, UK) and thermometer (CheckTemp, Hanna Instruments, Rhode Island, USA), respectively.

Pre-trial standardisations

Participants recorded all food and drink consumed within 24 hours (including nutritive supplements, e.g., whey protein), and exercise performed within 48 hours, before their first experimental condition (abstaining from performing any exercise or other strenuous physical activity within 24 hours). Participants drank at least 40 mL/kg body mass of fluid and refrained from consuming alcohol the day before experimental conditions. Both exercise and diet records

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were replicated before the second condition. The night before each lab visit, participants finished eating and drinking by 22:00 and spent at least seven hours in bed at night, arriving fasted the following morning. At least one week washout was required before experimental conditions for any participant who had travelled abroad.

Study protocol

Adherence to pre-trial standardisations was confirmed via questionnaire upon arrival at the laboratory (8:00-9:30). A baseline venous blood sample (6 mL) was collected after 15 min seated rest via venepuncture of an antecubital vein, then breakfast was provided. Breakfast was consumed within 10 min and consisted of flapjack and orange juice (28% fat, 67% carbohydrate, 5% protein) from Tesco PLC (Welwyn Garden City, UK). Energy content was 20% of estimated daily energy requirements (38) multiplied by a physical activity factor of 1.8 (644 ± 35 kcal, 20.0 ± 1.1 g fat, 107.5 ± 5.8 g carbohydrate, 8.5 ± 0.5 g protein).

Participants received four capsules (Licaps®, Lonza, Basel, Switzerland) 15 min after starting breakfast, each containing 0.5 mL hemp oil (Bridge Farm Nurseries ltd, Spalding, UK) at 0 (placebo) or 149 mg/mL CBD isolate (CBD; 298 mg total; KannaSwiss, Kölliken, Switzerland). CBD concentration and absence of quantifiable THC concentrations were confirmed via high performance liquid chromatography. This dose was selected in line with studies that have demonstrated physiological/psychological effects following acute CBD ingestion (300-600 mg) (24,39) but is greater than current consumer-recommended doses in some nations, e.g., 150 mg/day in Australia and 10 mg/day in the UK.

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After supplementation, participants remained resting for 90 min, during which time wireless thermistors (iButton DS1922L Thermochron Data Logger, Berkshire, UK) were fitted to the skin surface using kinesiology tape over the biceps brachii, chest, calf and quadriceps (40), to obtain weighted mean skin temperature (41). If a thermistor became unattached during exercise, the formula was adjusted to maintain the same relative weighting and applied to both trials for that participant. Five, one and one participants' data was obtained from three, two and one thermistor(s), respectively. During the final 5 min of rest, core temperature was recorded, subjective measurements completed, and a blood sample obtained. Core temperature was measured using a radiotelemetry pill (e-Celsius Performance, Hérouville Saint-Clair, France) ingested immediately before going to bed the night beforehand. Subjective measurements were attained using visual Likert-type scales for feeling (-5 to +5; 'very good' to 'very bad') (42), arousal (1 to 6; 'low' to 'high') (43), thermal comfort (-10 to +10; 'unbearably cold' to 'unbearably hot') (44), thirst (0 to 10; 'not at all' to 'very') (45) and gastrointestinal discomfort (0 to 10; 'very comfortable' to 'extremely uncomfortable') (45). After this, nude body mass was measured. Participants then began exercise 2 h after beginning breakfast (105 minutes after supplementation).

Treadmill exercise was 60 min at a constant speed corresponding to 60-65% $\dot{V}O_2$peak, in 32°C and 50% rh, with a front-facing fan (wind speed ~4.7 km/h). Core temperature, skin temperature and heart rate were recorded at 16, 36 and 56 minutes (at 15-second intervals, averaged over one min). Heart rate was measured using a telemetric monitor (Garmin HRM1G, Olathe, USA). The subjective measurements described above and perceived exertion (RPE; 6-

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20) (46) were recorded at 18, 38 and 58 minutes. During exercise, participants were allowed to listen to a standard music playlist that was repeated in both conditions.

Immediately following exercise, participants towel-dried and nude body mass was recorded. Following this, the Physical Activity Enjoyment Scale (PACES) (47) and subjective measurements were completed. The PACES assesses enjoyment for physical activity using 7-point bipolar Likert scales (scored 18 to 126; greater scores indicate greater enjoyment). After this, participants rested in a seated position for 15 min until a blood sample was collected (20 min post-exercise). Participants then remained resting, receiving flapjack and orange juice 45 minutes post-exercise (same nutrient content as breakfast; consumed within 15 minutes). A final blood sample was obtained 90 min post-exercise. Following the second condition, participants completed an exit questionnaire regarding study blinding. They were asked if they believed they could correctly identify which trial was placebo/CBD ('yes'/'no'), and to identify which trials were placebo/CBD.

Blood Samples

EDTA-treated whole-blood samples (1 mL) were analysed for haemoglobin (Hb) and haematocrit concentrations. Hb concentration was assessed in duplicate via the cyanmethemoglobin method (48), with absorbance measured at 450 nm using a Novaspec II spectrophotometer (Amersham Pharmacia Biotech Inc, Piscataway, USA). Haematocrit was assessed in triplicate via microcentrifugation (mylab HC-01, AHN Biotechnologie GmbH, Nordhausen, Germany) and a micro-haematocrit reader (Hawksley, Lancing, UK). These values were used to calculate plasma volume changes relative to baseline blood samples (48).

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Remaining EDTA-treated and Lithium heparin-treated whole-blood samples (5 mL each) were centrifuged (10 minutes, 2500 rcf, 4°C) and the plasma stored at -80°C until analysis. IL-6, IL-8 and TNF-α were analysed in EDTA-treated plasma via an automated enzyme linked-immunosorbent assay (ELISA; ST01A-PS-003229, ELLA™, ProteinSimple, Bio-Techne, Oxford, UK). Intra-assay coefficient of variation (CV) was 4.2%, 2.6% and 1.5%, respectively. Lithium-heparin-treated plasma was analysed for I-FABP and CD14 via ELISA (HK406, Hyult Biotech, Wayne, USA; DC140, R&D Systems, Minneapolis, USA), osmolality via freezing point depression (Gonotec Osmomat 030 Cryoscopic Osmometer, Gonotec, Berlin, Germany) and CBD via liquid chromatography-mass spectrometry (LC-MS). Intra-assay CV was 14.6%, 4.8%, 0.3% and 7.5%, respectively.

CBD was quantified with a validated method (49) using an Acquity UPLC (Waters Corp., MA, U.S) coupled to a Quattro Ultima Mass Spectrometer (Micromass, now Waters Corp.), with MassLynx software version 4.1 (Waters Corp.). Plasma samples (200 µL) were deproteinised using labelled internal standard (200 µL of 120 ng/mL CBD-d3 in methanol; Cerilliant®, TX, USA) and acetonitrile (400 µL), then vortexed vigorously for 30 seconds. Proteins were precipitated via centrifugation (21.1 rcf, 15 minutes), then the supernatant dried via vacuum centrifugation before reconstitution in 100 µL of 60:40 methanol:0.1% formic acid in water. Chromatography was performed using a Zorbax XDB-C18 column (50 x 2.1 mm, 3.5 µm particle size; Agilent, CA, USA), coupled to an Agilent XDB-C8 guard column (12.5 x 2.1 mm, 3.5 µm particle size), with mobile buffers of 0.1% formic acid in water (A) and methanol (B). A 5-minute gradient protocol was employed (column temperature 50°C; flow rate 600 µL/min), which started at 60% B, held for 0.5 min, increased to 69% B at 1.9 min, then 90% B at

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2 min, held until 3 min, increased to 100% B at 3.1 min and held until re-equilibration at 60% B after 4 min. A 20 µL injection was used in full loop mode with an overfill factor of 1.0. CBD and CBD-d3 were analysed via positive electrospray ionisation in multiple reaction monitoring mode. Analyte transitions of *m/z* 315.1>193.1 (collision energy 29 eV, dwell time 100 ms) and *m/z* 318.1>196.1 (20 eV, 100 ms) were monitored for CBD and CBD-d3, respectively. MS source settings included: capillary voltage 3 kV, cone voltage 35 V, source temperature 150°C, desolvation gas 400°C at 500 L/h, and cone gas 200 L/h. Analyte peak areas were integrated using QuanLynx (Waters Corp.) and the peak area of CBD was calculated against the peak area of CBD-d3. CBD concentrations were determined by comparison to a calibration curve obtained using plasma spiked with CBD in methanol (Cerilliant®) to obtain concentrations between 5 and 300 ng/mL. Concentrations below or above the standard curve were assigned upper and lower concentration limits, respectively. Intra-assay CV was 7.5%.

Statistical Analysis

Condition averages are reported as mean ± standard deviation (SD) and between-conditions differences (calculated as CBD-placebo) reported as mean (95% confidence intervals [CIs]) when normally distributed. Skewed data are presented as median (interquartile range). Data were screened for outliers using criteria of ±3SD and ±3MAD (median absolute difference) (50) and reported with and without outliers when applicable. Area under the curve (AUC) was calculated for CBD and inflammatory cytokines (to give an indication of total systemic response) using the trapezoid rule via an automated spreadsheet (51). Relationships between peak CBD concentrations, CBD AUC, and between-conditions differences for mean core temperature, mean skin temperature, sweat loss, peak/AUC values for each inflammatory cytokine, and CD14 90

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minutes post-exercise were assessed using Pearson correlations or Spearman's rank, when data were normally distributed (Shapiro Wilk test $p > 0.05$) or not, respectively.

Between-conditions comparisons were made via paired *t*-test when data were normally distributed, and via Wilcoxon-Signed Rank test when not. For time-series data, two-way repeated measures ANOVA was used, with condition and time as within-subjects factors. A Greenhouse-Geisser correction was applied where sphericity was violated ($\varepsilon < 0.75$) and Huynh-Feldt correction applied otherwise. If $p \leq 0.05$ for a condition*time interaction, *post hoc* comparisons were made via paired *t*-tests or Wilcoxon-Signed Rank tests, with a Holm-Bonferroni correction. Mean core temperature, mean skin temperature, sweat loss, and AUC for inflammatory cytokines were assessed for effects of trial order via paired *t*-test or Wilcoxon-Signed Rank test, depending on the normality of data distributions ($p > 0.05$ unless reported). Statistical analyses were conducted in SPSS 27 (IBM Corp, Armonk, USA), with statistical significance accepted at $p \leq 0.05$.

RESULTS

Physiological and perceptual responses during exercise

Heart rate, core temperature and skin temperature increased throughout exercise (time $p < 0.001$; Figure 1), but with no condition*time interactions ($p \geq 0.180$). RPE, thermal comfort, thirst and arousal also increased throughout exercise (time $p \leq 0.014$), whereas gastrointestinal comfort did not change (time $p = 0.053$) and feeling decreased (time $p = 0.03$; Figure 2). There were also no condition*time interactions ($p \leq 0.254$; Figure 2) or between-conditions differences in mean values throughout exercise for those outcomes ($p \geq 0.059$; Table 1). Sweat loss during

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exercise and physical activity enjoyment did not differ between conditions ($p \geq 0.718$; Table 1) and there was no condition*time interaction for plasma osmolality ($p = 0.711$) or plasma volume change ($p = 0.114$; Figure 3).

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Intestinal Damage and Inflammatory Markers

Plasma I-FABP decreased between baseline and exercise across both conditions, then increased, returning to baseline values after exercise (time $p = 0.002$), but there was no condition*time interaction ($p = 0.101$; Figure 3). However, there was an order effect for I-FABP AUC, whereby values were (median) 17935 (5600, 41959) pg/mL greater in the second condition ($p = 0.013$). There was a condition*time interaction for CD14 ($p = 0.030$), with concentrations diverging between conditions 90 minutes after exercise, such that CD14 was 1776 (463 to 3090) pg/mL less in CBD than placebo ($p = 0.049$; Figure 3).

Each inflammatory cytokine increased after exercise (time $p \leq 0.002$; Figure 4). Two outliers were identified for IL-6 (Figure 4); one individual had baseline IL-6 concentrations above 7 pg/mL (>3SD and >3MAD above population means) and one had an exaggerated post-exercise response (paired differences in post-exercise concentrations and AUC >3SD and >3MAD). Including outliers, median differences were -113 (-172, 27) pg/mL·270min for IL-6 AUC ($p = 0.054$) and -0.8 (-1.1, -0.3) pg/mL for IL-6 C_{max} ($p = 0.050$). Excluding outliers (n = 11), IL-6 AUC and C_{max} were 89 (26 to 151) pg/mL·270min ($p = 0.013$) and 0.62 (0.22 to 1.02) pg/mL ($p < 0.001$) lower in CBD than placebo. However, there was an order effect for IL-6 AUC when outliers were excluded, whereby the response was 73.5 (1.6 to 145.4) pg/mL·270min lower

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in the second condition (p = 0.046). There was no order effect for AUC with the full dataset (p = 0.134). There was no condition*time interaction for IL-6 concentrations with (p = 0.180) or without (p = 0.216) outliers included.

There was no condition*time interaction or between-conditions differences in C_{max} or AUC for IL-8 (Figure 4; Table 1). There was an interaction effect for TNF-α ($p < 0.001$), whereby concentrations decreased between baseline and exercise in both conditions ($p < 0.001$), but to a greater extent in placebo (mean difference 15 min pre-exercise: 0.58 [0.18 to 0.99] pg/mL; p = 0.035). Concentrations rebounded after exercise in both conditions, but to a greater extent in placebo, such that concentrations 20 minutes after exercise increased above baseline in placebo (p = 0.003) but not CBD (p = 0.909; Figure 4). The between-conditions difference at this point was 0.42 (0.00 to 0.84) pg/mL (p = 0.143). TNF-α concentrations were not different to baseline in both conditions 90 minutes after exercise ($p \geq 0.155$). There was no between-conditions difference in TNF-α AUC (p = 0.614) or C_{max} (p = 0.116).

Cannabidiol

Plasma CBD C_{max} was 190 ± 59 ng/mL, occurring after 270 minutes (90 minutes post-exercise) for all participants (Figure 5). CBD AUC was 18684 ± 11583 ng/mL·270min. There was a positive relationship between peak CBD concentration and the between conditions difference in mean core temperature during exercise (r = 0.67; p = 0.013), but no relationships between other outcomes. No CBD was detected during the placebo condition.

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Study Blinding

No participant believed that they could identify which trials were CBD/placebo, and 7/13 were correct when asked to guess the trial order.

DISCUSSION

CBD ingestion influenced the immune response to one hour of treadmill exercise in 32°C and 50% rh compared to placebo, with lower plasma CD14 90-minutes after exercise and reduced peak IL-6 concentration. The magnitudes of these differences were trivial, and as such, CBD did not influence the increase in core temperature or other thermoregulatory responses to exertional heat stress (increased skin temperature, sweat loss, and reduced thermal comfort). Other physiological and subjective responses to exertional heat stress were also unaffected by CBD ingestion, including increases in heart rate and perceived exertion. This is one of the first studies to indicate that CBD may influence immune function in humans (24,52), whereby anti-inflammatory effects have predominantly been demonstrated in preclinical models (53). Therefore, it remains possible that CBD could have clinical utility in other inflammatory conditions.

Plasma CD14 increased 90 minutes after exercise in placebo relative to CBD. Increases in plasma CD14 during exertional heat stress indicate monocyte activation by bacterial components, such as LPS, that leak into circulation as intestinal integrity is compromised (7,16). Therefore, there are two potential explanations as to how CBD could have reduced CD14 concentration. First, that CBD was protective against intestinal ischemic damage, such that subsequent systemic exposure to LPS was reduced. Second, that LPS exposure was equivalent

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between conditions, but CBD attenuated CD14 release by LPS-exposed monocytes. Both are plausible based on pre-clinical evidence (20–23,30,31), but the capacity to draw conclusions about the degree of intestinal damage from plasma I-FABP concentrations was limited by the presence of a trial-order effect for I-FABP AUC. It is not clear why intestinal damage may have increased from the first to second condition, but a longer washout period between trials may have prevented that occurring. The difference in CD14 coincided with peak plasma CBD concentrations, supporting the plausibility of an effect of CBD (via either described mechanism).

Notwithstanding this trial-order effect for I-FABP, the magnitude of intestinal damage was not substantial in either condition, which likely limited the degree of any endotoxin release and subsequent pro-inflammatory effects. Post-exercise I-FABP increased by 459 and 305 ng/mL from pre-to-post-exercise but did not increase above baseline concentrations, and there were no changes in GI discomfort throughout exercise. Longer bouts of running (≥2 hr) in greater ambient temperature (e.g., 35°C) in well trained participants have induced changes in I-FABP >1000 pg/mL, which have been associated with gastrointestinal discomfort (54). Therefore, similar investigations in the future would benefit using a longer or more intense exertional heat stress model to induce a greater disruption to intestinal integrity (10). A shorter duration was used in the present investigation as participants were recreationally active. Less trained individuals demonstrate greater intestinal permeability and lower tolerance to a given increase in core temperature (55,56). However, over half of participants did not reach core temperatures of over 39°C, which has been recommended to induce significant gastrointestinal barrier disruption (57). Furthermore, recent work suggests that ingestion of carbohydrate or amino acids before and during exertional heat stress suppresses the ensuing intestinal damage

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(58,59). Hence, the consumption of breakfast may explain the decrease in I-FABP between baseline and pre-exercise. (58,59). Future studies that measure I-FABP should consider potential interactions with diet.

Attenuated monocyte activation, as indicated by lower post-exercise CD14, could have contributed to the 8-12% reduction in peak IL-6 concentration in the CBD condition. Monocyte activation elicits secretion of inflammatory cytokines (14,16), and, in line with this, IL-6, IL-8 and TNF-α did increase post-exercise. Alternatively, reduced IL-6 concentrations could be indicative of attenuated secretion by other immune cells (e.g., neutrophils) (15) or skeletal-muscle (60). If CBD influenced IL-6 secretion from monocytes alone, this could explain why only a marginal reduction was seen, as a substantial portion of circulating IL-6 after exertional heat stress is likely muscle-derived (61). Regulation of IL-6 secretion from skeletal muscle is not well understood (62), making it difficult to speculate on whether and how CBD may have influenced this.

Potentially counter to the hypothesised reduction in monocyte activation, was the absence of a reduction in IL-8 and TNF- α following CBD ingestion. IL-8 did not differ between conditions, whereas TNF-α decreased before exercise in placebo relative to CBD. This could be viewed as a pro-inflammatory effect of CBD, i.e., preventing a reduction in TNF-α before exercise. It is not clear why TNF-α would decrease during this resting period, other than in relation to diurnal rhythmicity (63). TNF-α concentrations did rebound in placebo, such that total TNF-α exposure was not different between conditions, and TNF-α increased above baseline in only the placebo condition. The latter of which, could be viewed as an anti-inflammatory effect,

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whereby CBD attenuated the post-exercise increase in TNF-α above baseline that was only seen in the placebo condition. These findings are difficult to interpret but are likely inconsequential in the present scenario as the TNF-α increase above baseline was minimal throughout the study.

Only one other randomised controlled trial has observed reductions in circulating IL-6 after CBD administration compared to placebo, with these observations made after 12 weeks of 800 mg/day supplementation in 48 people with cocaine use disorder (52). Therefore, the present study is the first to indicate that *acute* CBD ingestion can reduce the IL-6 response in humans. Whilst the magnitude of difference in peak IL-6 concentration observed in this investigation (0.6-0.8 pg/mL) was not physiologically meaningful, it must be noted that this difference was observed in the context of a modest total increase in IL-6. The peak IL-6 concentrations seen in this investigation (~5 pg/mL) are not much greater than those typically seen at rest in healthy populations aged <30 years (2-4 pg/mL) (64), whereas mean concentrations of >10 pg/mL can be seen after more prolonged exertional heat stress (≥2 hr) in lab settings (54) and values >40 pg/mL are consistently seen in cases of heat stroke (65). Therefore, it remains possible that CBD ingestion could induce a greater magnitude of effect during a more pronounced inflammatory response. Hence, further investigation under conditions of more prolonged or severe heat stress is likely warranted, and clinical utility of CBD in other pro-inflammatory conditions remains possible. Follow-up research may also benefit from more frequent blood sampling, to better capture the time course of responses, as well as a greater washout period between conditions to mitigate against order effects, such as was seen with the IL-6 AUC when outliers were excluded.

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There were no effects of CBD on sweat loss during exercise or hydration markers across trials, nor on any subjective outcome. CBD can elicit acute psychological effects, but these have been limited to outcomes like anxiety at present (66). The correlational direction between peak CBD concentrations and between-conditions differences in mean core temperature during exercise was opposite to what would be expected if greater CBD concentrations were exerting an effect on core temperature, i.e., those with a greater peak CBD concentration were more likely to have a greater core temperature in the CBD condition. However, little can be drawn from this at present, seeing as CBD was substantially elevated in all participants and peak CBD concentrations occurred 90 minutes after exercise. [NO_PRINTED_FORM]It must also be noted that the present findings cannot necessarily be translated to female populations, and further research may seek to examine more well-trained athletes who can sustain more prolonged exercise with greater perturbations in core temperature (55).

It is not clear why CBD did not appear in plasma at the onset of exercise (105 min after dosing) as CBD typically peaks between 1-3 hours in the fed state (55). Information is scarce regarding the interaction between various dietary factors and CBD pharmacokinetics but requires further exploration and may help explain the present observations. This absence of plasma CBD at exercise onset may have limited the CBD concentrations achieved later in exercise, when core temperature was greatest. There was a wide range of CBD concentrations at exercise cessation, with several individuals displaying comparatively low plasma CBD concentrations to others, and one individual showing no detectable increase. This variability is typical of CBD pharmacokinetics and represents a great challenge for eliciting acute responses (55). However, it is speculative whether greater CBD concentrations would have enhanced CBD-induced effects at

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a population level, as therapeutic plasma CBD ranges have not been established and CBD can show an inverted U-shaped dose response (67). The peak in CBD concentrations 90 min after exercise may explain why a difference between conditions in CD14 was only seen at this timepoint[NO_PRINTED_FORM][NO_PRINTED_FORM]

Overall, CBD ingestion did not improve thermoregulation during exercise in a hot environment but induced minor reductions in plasma IL-6 and CD14. The potential for CBD to attenuate inflammatory responses, even when heat strain and gastrointestinal damage were not particularly pronounced, warrants further investigation under more severe or prolonged heat stress. This is pertinent considering that attenuating the inflammatory response during exertional heat stress could have benefits independent from thermoregulation, such as mitigating the adverse consequences of heat illness. Furthermore, the observation of attenuated immune responses *in vivo* in humans is an important step in translating the substantial preclinical evidence base toward potential therapeutic use for CBD.

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Data Availability: The datasets generated during and/or analysed during the current study are available from the corresponding author upon reasonable request.

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FIGURE LEGENDS

Figure 1. Physiological variables measured during exercise in CBD and placebo conditions; time begins relative to starting exercise; 1A Core Temperature (condition $p = 0.083$, time $p < 0.001$, interaction $p = 0.396$); 1B Peak Core Temperature; 1C Skin Temperature (condition $p = 0.344$, time $p < 0.001$, interaction $p = 0.761$); 1D Heart Rate (HR) (condition $p = 0.259$, time $p < 0.001$, interaction $p = 0.368$).

Figure 2. Subjective variables measured during exercise in CBD and placebo conditions; time begins relative to starting exercise; 2A: Rating of Perceived Exertion (RPE) (condition $p = 0.759$, time $p < 0.001$, interaction $p = 0.316$); 2B: Thermal Comfort (condition $p = 0.666$, time $p < 0.001$, interaction $p = 0.664$); 2C: Thirst (condition $p = 0.614$, time $p < 0.001$, interaction $p = 0.559$); 2D: Gastrointestinal (GI) Discomfort (condition $p = 0.400$, time $p = 0.053$, interaction $p = 0.261$); 2E: Feeling (condition $p = 0.016$, time $p = 0.001$, interaction $p = 0.735$); 2F: Arousal (condition $p = 0.818$, time $p = 0.014$, interaction $p = 0.254$).

Figure 3. Blood-based markers of hydration, and plasma intestinal fatty acid binding protein (I-FABP) and soluble cluster of differentiation 14 (CD14) across CBD and placebo conditions; values and error bars are mean ± standard deviation; time begins relative to starting breakfast, and exercise was performed at 120-180 minutes; 3A: Plasma Osmolality (condition $p = 0.506$, time $p < 0.001$, interaction $p = 0.711$); 3B: Plasma Volume (condition $p = 0.858$, time $p = 0.047$, interaction $p = 0.114$); 3C: Plasma I-FABP (condition $p = 0.685$, time $p = 0.002$, interaction $p = 0.101$; 3D: Plasma I-FABP area under curve; 3E: Plasma CD14 (condition $p = 0.372$, time $p =$

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0.500, interaction p = 0.030); 3F: Plasma CD14 area under curve; * denotes $p \leq 0.05$ between conditions.

Figure 4. Plasma concentration of inflammatory cytokines and their respective area under curve (AUC) across CBD and placebo conditions; values and error bars are mean ± standard deviation; time begins relative to starting breakfast, and exercise was performed at 120-180 minutes; 4A: IL-6 (n = 13) (condition p = 0.333, time p = 0.003, interaction p = 0.180); 4B: IL-6 (n = 13) AUC; 4C: IL-6 (n = 11) (condition p = 0.015, time $p < 0.001$, interaction p = 0.216); 4D: IL-6 (n = 11) AUC; 4E: IL-8 (condition p = 0.823, time $p < 0.001$, interaction p = 0.921); 4F: IL-8 AUC; 4G: TNF-α (condition p = 0.888, time $p < 0.001$, interaction $p < 0.001$); 4H: TNF-α AUC; † denotes $p \leq 0.05$ against baseline within CBD; ‡ denotes $p \leq 0.05$ against baseline within placebo; * denotes $p \leq 0.05$ between conditions; p values above AUCs relate to between-condition comparisons.

Figure 5. Plasma CBD concentrations across the CBD condition; values and error bars are mean ± standard deviation; grey line represent individual responses; time begins relative to starting breakfast, and exercise was performed at 120-180 minutes.

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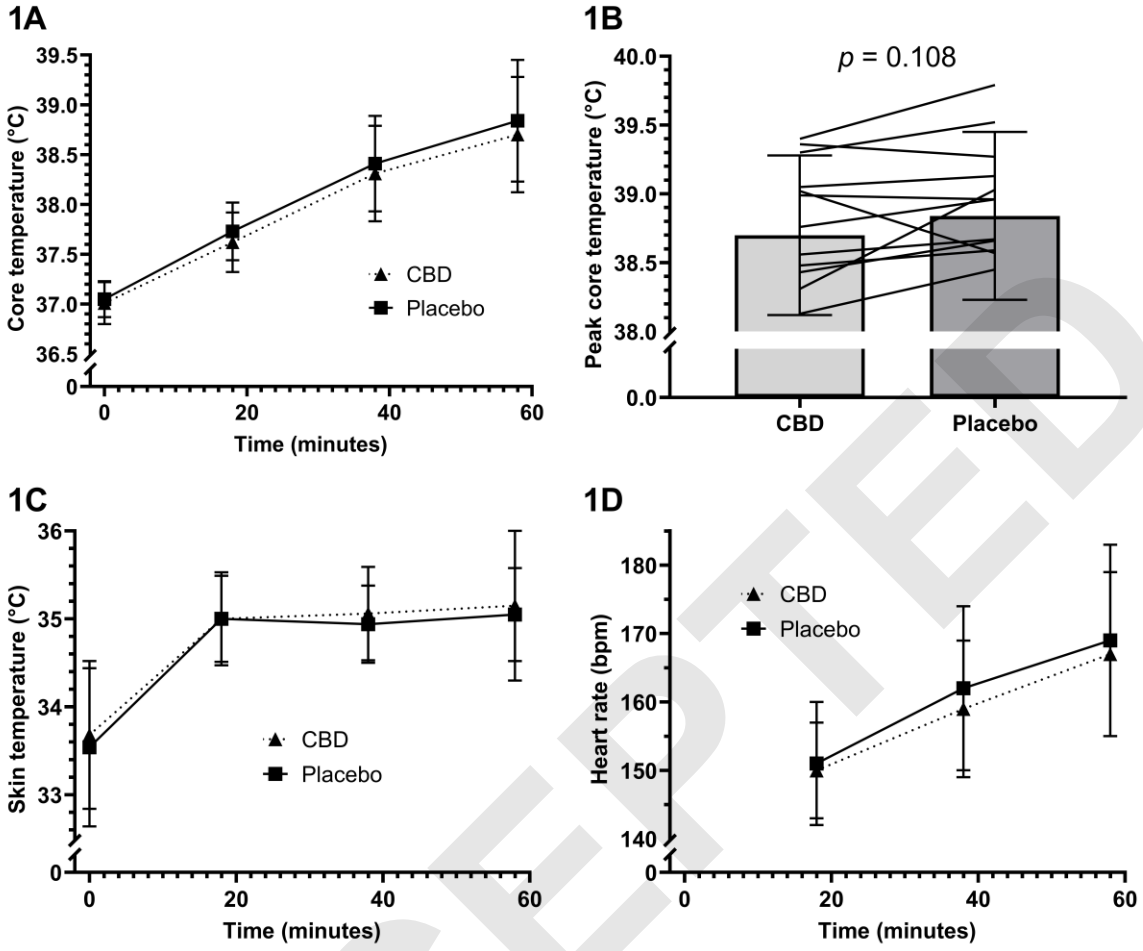

Figure 1

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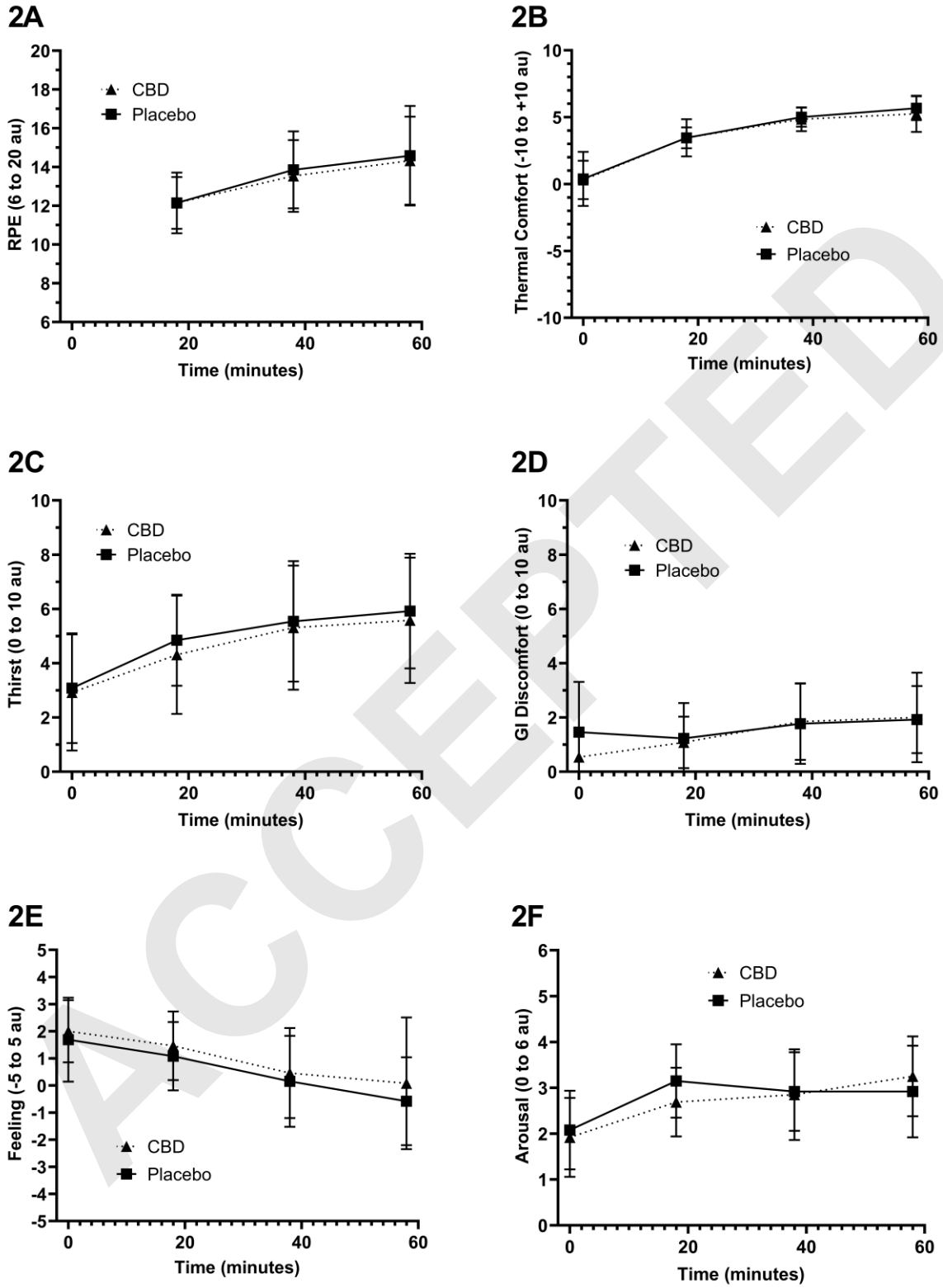

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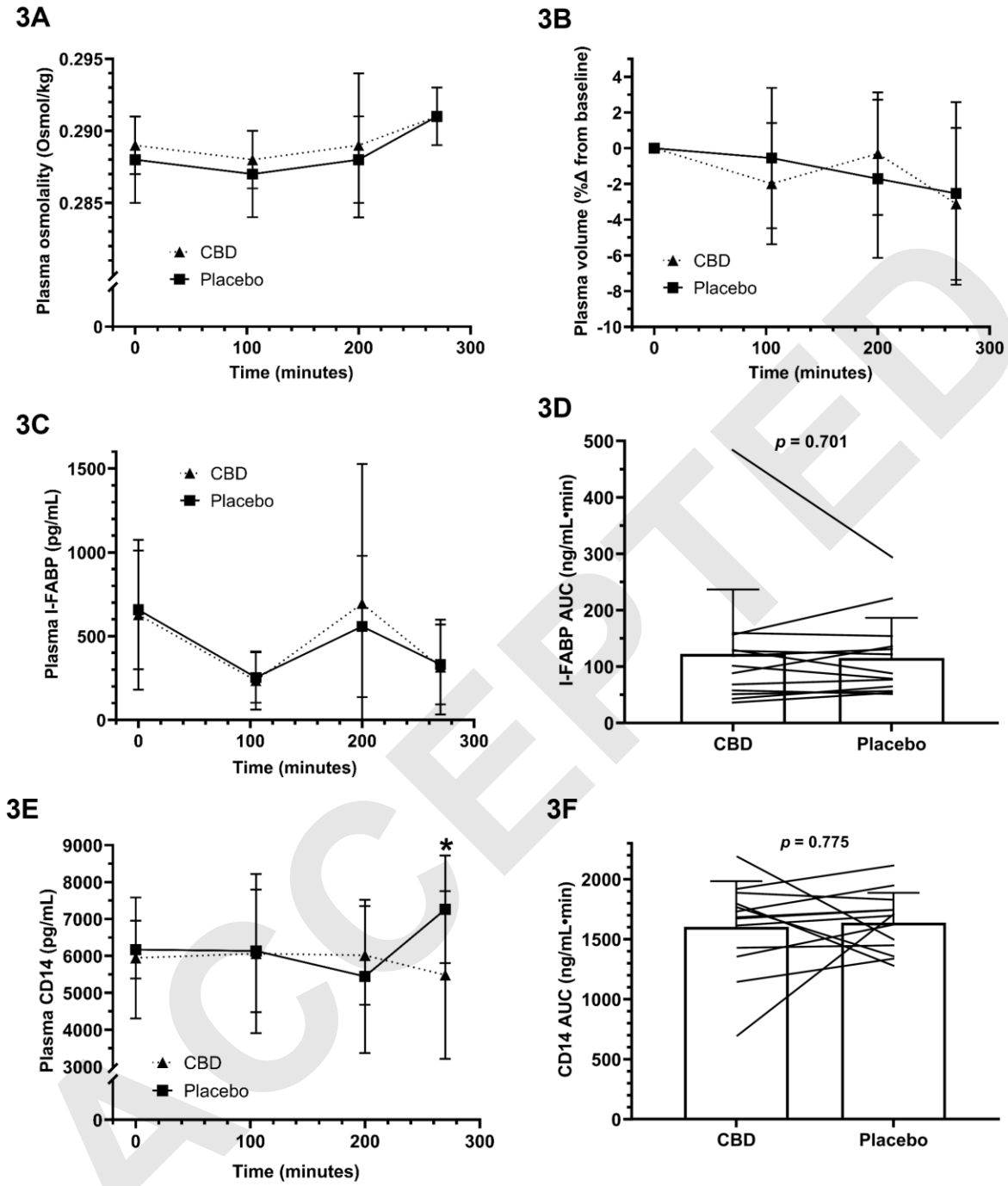

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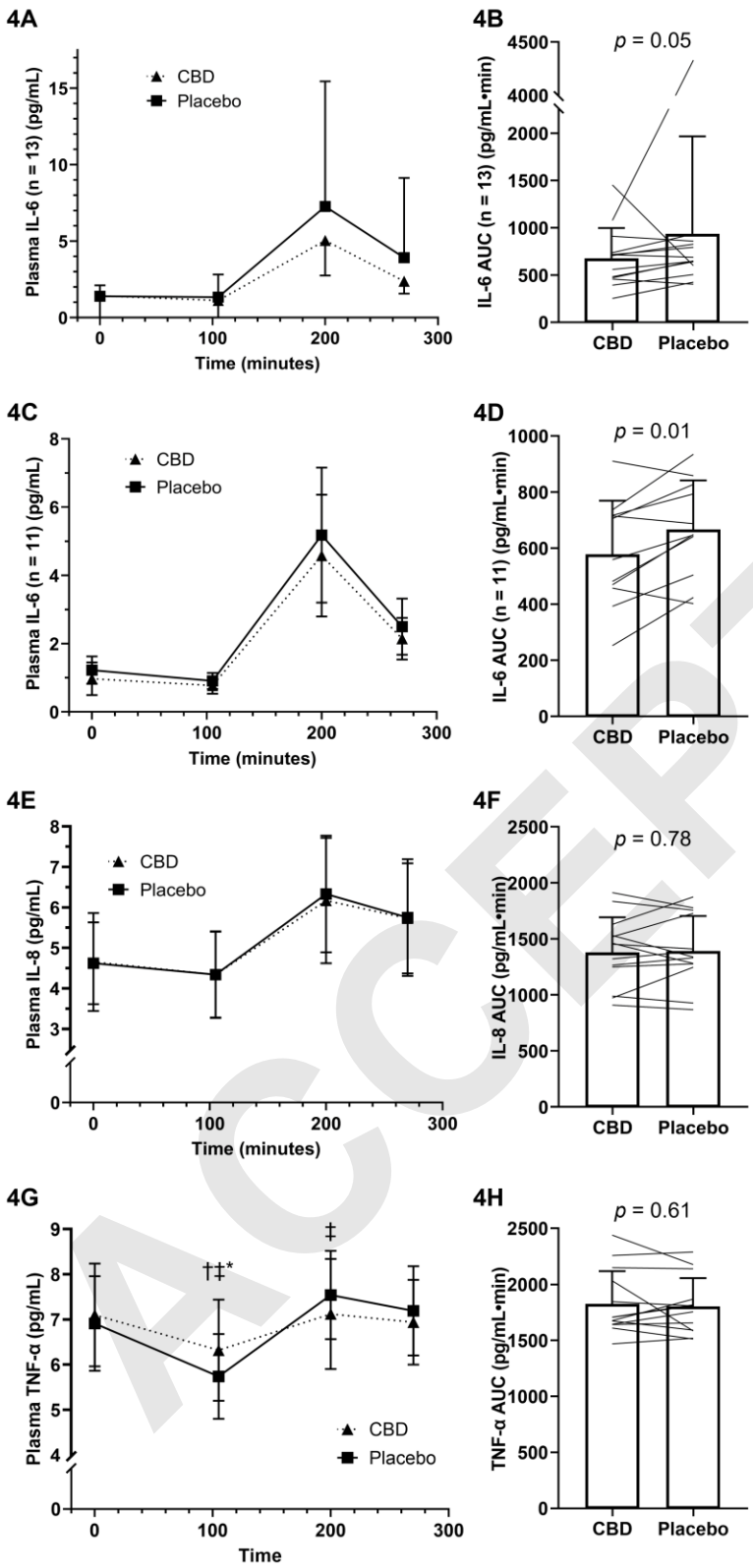

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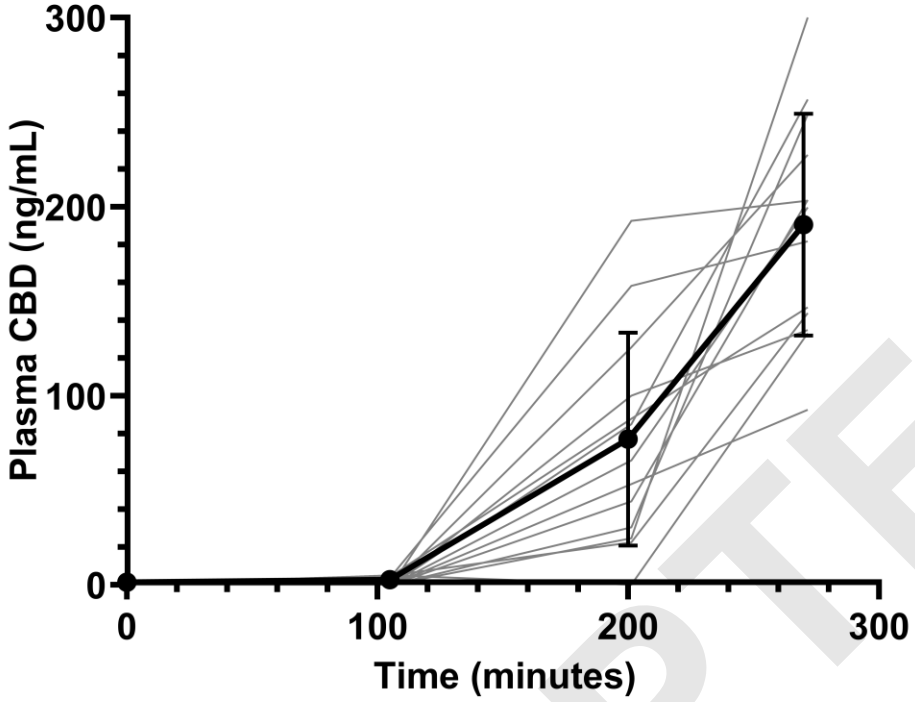

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Table 1. Mean values for outcomes during exercise, physical activity enjoyment (PACES) and peak concentrations of inflammatory cytokines.

	CBD	**Placebo**	**Difference**	***p* value**
Mean core temperature (°C)	38.21 ± 0.44	38.33 ± 0.44	-0.10 (-0.25 to 0.02)	0.074
Peak core temperature (°C)	38.70 ± 0.58	38.84 ± 0.61	-0.13 (-0.30 to 0.03)	0.108
Δ core temperature (°C)	1.69 ± 0.48	1.79 ± 0.53	-0.10 (-0.27 to 0.08)	-
Mean skin temperature (°C)	34.72 ± 0.46	34.63 ± 0.39	0.09 (-0.11 to 0.29)	0.345
Mean heart rate (bpm)	159 ± 9	161 ± 11	-1 (-4 to 1)	0.265
Sweat loss (kg)	1.16 ± 0.23	1.15 ± 0.19	0.01 (-0.05 to 0.06)	0.718
Mean RPE (6 to 20 au)	13 ± 2	13 ± 2	0 (-1 to 1)	0.483
Mean feeling (-5 to +5 au)	1 ± 2	0 ± 1	0 (0 to 1)	0.059
Mean arousal (1 to 6 au)	3 ± 1	3 ± 1	0 (-1 to 0)	0.549
Mean thermal comfort (-10 to +10)	5 ± 1	5 ± 1	0 (-1 to 0)	0.303
Mean thirst (0 to 10 au)	5 ± 2	5 ± 2	0 (-1 to 0)	0.388
Mean GI discomfort (0 to 10 au)	2 ± 1	2 ± 1	0 (-1 to 1)	0.656
PA enjoyment	72 ± 5	73 ± 4	-1 (-4 to 2)	0.476

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(18 to 126 au)				
IL-6 C_{max} (n = 13) (ng/mL)	5.2 ± 2.4	7.3 ± 8.2	-0.8 (-1.1, -0.3) [a]	0.050
IL-6 C_{max} (n = 11) (ng/mL)	4.6 ± 1.8	5.2 ± 1.8	-0.6 (-1.0 to -0.2)	< 0.001
IL-8 C_{max} (ng/mL)	6.3 ± 1.5	6.4 ± 1.4	-0.0 (-0.7 to 0.7)	0.915
TNF-α C_{max} (ng/mL)	7.4 ± 1.2	7.6 ± 1.0	-0.2 (-0.7 to 0.4)	0.116

Condition summary statistics are mean ± standard deviation; differences between conditions are mean (95% confidence intervals); RPE = rating of perceived exertion; GI = gastrointestinal; PA = physical activity.

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